

A cartoon character with a large blue question mark above its head. The character has a white face with large, expressive eyes and a small black mustache. It is wearing a blue suit jacket over a white shirt and a pink tie. The character is standing on a grey base.

Are You In Suspense?

Are you wondering what happened to your order or payment, even after a fortnight? But did you mention your name and address clearly while sending payment, and what the payment was for?

We get several payments every month without adequate instructions or even identity of the person sending the payment. As a result we are unable to take appropriate action on these, till we get a complaint - if at all.

At EFY, we want you to be our satisfied customer. So please do let us know in case your payment or instructions are not even acknowledged within a fortnight. We are not that lazy!

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of alternating dirt-cheap polystyrene that doesn't do anything, and 'responsive' poly-2 vinyl.

At rest, with no electrical charge, the screen is clear. As the voltage increases, poly-2 vinyl expands, becoming thicker and thicker while reflecting ever longer wavelengths of light, starting with blue and finishing with red at 10 volts.

The screen can also reflect non-visible wavelengths of light, such as infrared and ultraviolet, depending on the voltage applied. It can produce images using only a few volts because it doesn't create light, it only reflects it. In a dark room with no light, the screen would remain dark.

Chip-sized digital camera

A microchip-sized digital camera developed by the California Institute of Technology could provide vision for the US military's insect-sized aircrafts. The camera is so light that it can be carried by these tiny surveillance drones. It also uses very little power.

Today's minicams house image sensors and support circuitry on separate microchips, and most of the power is consumed in communication between the chips. Caltech's Jet Propulsion Lab has been able to squeeze all the components of a camera onto one low-power chip.

The gadget can be radio-controlled via a secure frequency-hopping link from up to a kilometre away, say the inventors.

Breathing batteries to store ten times the energy

Lithium-ion batteries used in laptops and cellphones are approaching their energy capacity limits. Researchers at University of St Andrews, the UK, claim to have found a way to break through their energy capacity barrier—let the batteries ‘breathe’ oxygen from the air.

Typically, a lithium-ion battery



An early demonstration model of the St Andrews air cell. Air enters and leaves via the porous circular membrane in the centre

contains a negative electrode of graphite, a positive electrode of lithium cobalt oxide and a lithium salt-containing electrolyte. Lithium ions move between the two electrodes during charging and discharging, sending electrons around the external circuit to power a gadget in the process. The major barrier to increasing the energy density of these batteries is the positive electrode. The lithium cobalt oxide is bulky and heavy. Researchers are trying to find a way to push up the amount of lithium stored there and thus the battery capacity.

The solution is to borrow the concept of zinc-air batteries used in hearing aids, which get their power-reacting zinc with oxygen from air. So UK researchers have begun designing a lithium-air battery. The new battery has a higher energy density than existing lithium-ion batteries because it no longer contains dense lithium cobalt oxide. Instead, the positive electrode is made from lightweight porous carbon, and the lithium ions are packed into the electrolyte which floods into the spongy material.

When the battery is discharged, oxygen from the air also floods through a membrane into the porous carbon, where it reacts with lithium ions in the electrolyte and electrons from the external circuit to form a solid lithium oxide.

The team's prototype device has a capacity-to-weight ratio of 4000 mAh per gram—eight times that of a cellphone battery. Even a ten times in-